

$$\begin{array}{ccc}
 \mathcal{I}(U) & \xrightarrow[\text{1-1}]{f} & f(\mathcal{I}(U)) \\
 \hat{\phi}_U \downarrow & & \downarrow \hat{\phi}_{f(U)} \\
 S^1 & \xrightarrow[\text{1-1}]{\hat{f}} & S^1 \\
 \text{id} \downarrow & & \downarrow \hat{f}^{-1} \\
 S^1 & \xrightarrow{\text{id}} & S^1
 \end{array}$$